

FOR THE COUNTY OF KAUAI

What the island tells us before it gets in the water.

Every day, thousands of visitors check Kauai's ocean conditions on their phones before they drive to a beach. That check is a stated intention, hours ahead of arrival. This document is a plain accounting of the 33 datasets that intention produces, ranked by what they are worth to the County — and a working dashboard that already renders them.

 **Aggregate only**  **Leading, not lagging**  **44 Kauai beaches live**

Prepared by Nick Mossman · OceanSafe (Tropical Agronomics dba) · oceansafety.app
Figures throughout are sample data unless a source is named.



THE PREMISE

The County measures what already happened. This measures what is about to.

Kaua'i's ocean-safety record is built from incident reports, and an incident report is a record of a failure that already occurred. Between the decision and the incident sits a gap of two to six hours in which a visitor picks a beach, looks up the conditions, and drives there. OceanSafe sits inside that gap — a free public-safety app running today across 44 Kaua'i beaches. Every check it serves is an early, voluntary, unprompted declaration of where a person intends to enter the water.

44

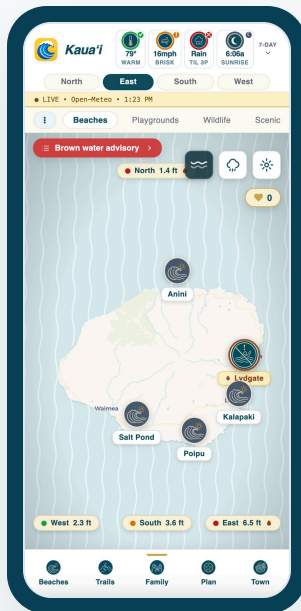
Kaua'i beaches with live conditions, hazards, and advisories in the app today

2–6 hrs

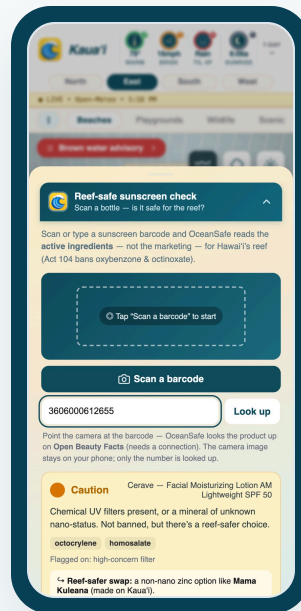
Typical window between a conditions check and arrival at the shoreline

0

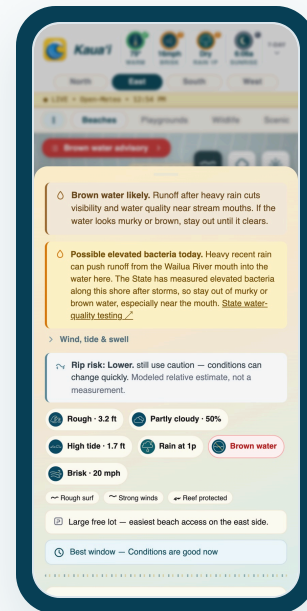
Individual records stored, reported, or sold — aggregates only, by design



Conditions today. Every shore scored and colour-coded — green, amber, red — with hazards flagged on the beach itself.



The reef scanner. A real barcode returning a real verdict — flagged on octocrylene and homosalate, with a Kaua'i-made swap offered.



The bacteria caution. Rain-keyed, tied to the Wailua River mouth, linking out to State water-quality testing.

All three are live in production today — real screens, captured from the public app, not mockups. Each one is also a measurement surface: what gets checked, what gets scanned, and whether a posted advisory was ever actually seen.



THE DASHBOARD · LIFE SAFETY

The number the County cannot currently produce.

A lifeguarded beach counts its own day; a tower is a census. Unguarded shoreline is the blind spot, and it is where Kaua'i's visitor drownings concentrate. The dashboard ranks every beach by how many people are planning to go there and colors the ones with no tower — four of the sample's six busiest sites have none.

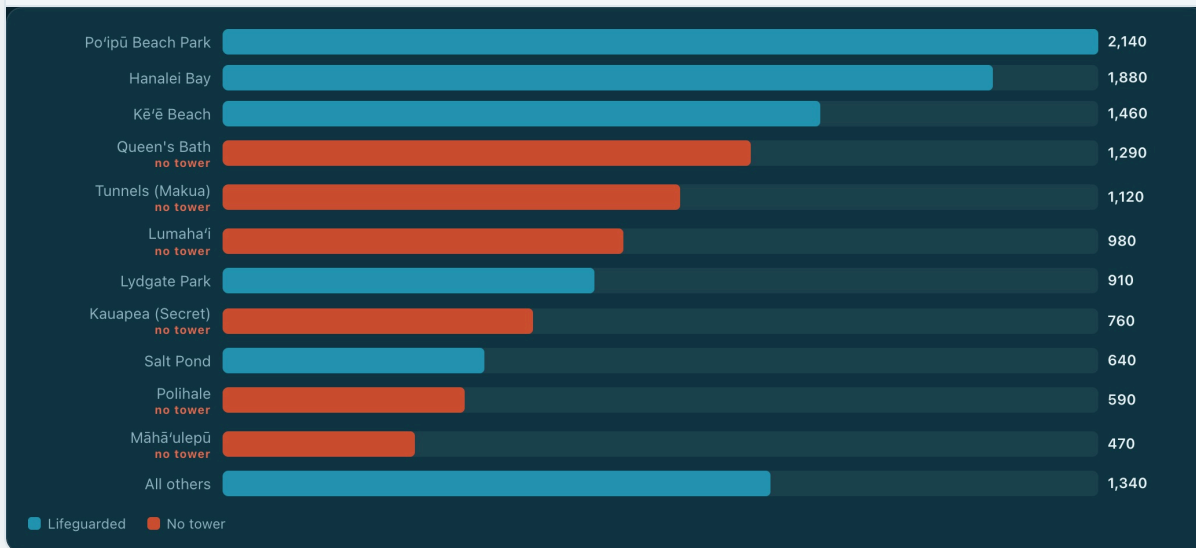
County dashboard — the headline figure

[SAMPLE DATA](#)

41%

of visitor beach checks point at
unguarded shoreline — the population
Ocean Safety cannot see until
something happens

Beach checks this month, ranked — guarded versus unguarded

[SAMPLE DATA](#)


Tower classifications are illustrative pending the County's authoritative roster. The value is the **ranking**: which unguarded sites are drawing the most people, refreshed weekly instead of reconstructed after an incident.

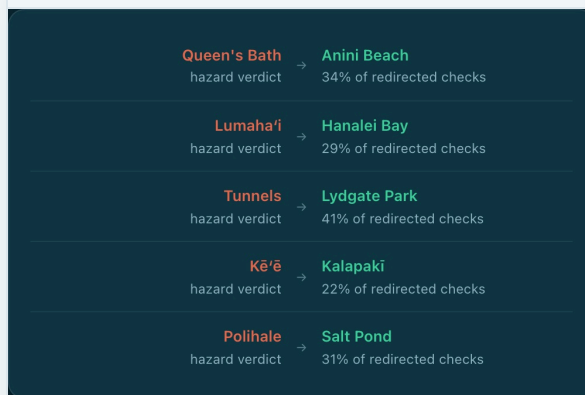


THE DASHBOARD · WARNING AND RESPONSE

Did the warning move anybody?

When the app returns a hazard verdict, some visitors change plans — and the next beach they check is the one that just inherited the crowd. No other County data source sees that substitution, and it says where the displaced people are about to arrive, hours before they do.

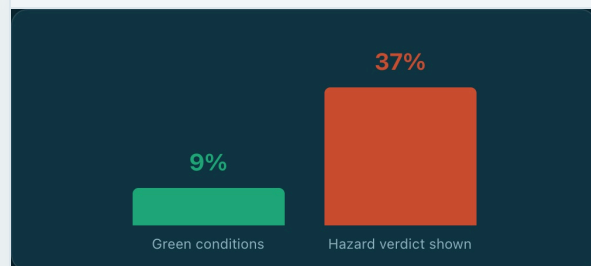
Hazard verdict → next beach checked

[SAMPLE DATA](#)


A north-shore hazard day pushes visitors east and south.

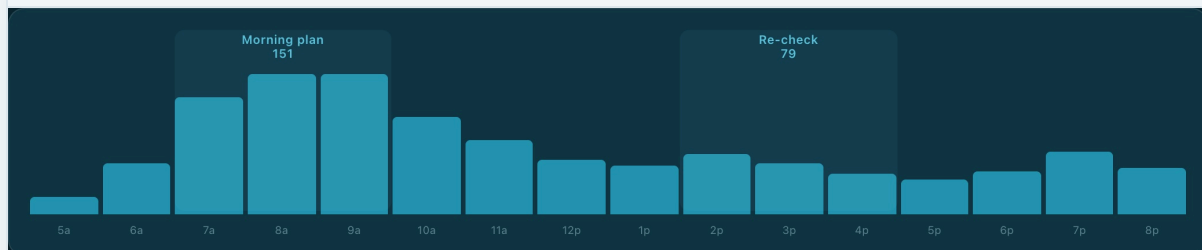
Lydgate absorbs the largest share — where the extra staffing and parking load actually lands.

Plan changed after a hazard verdict

[SAMPLE DATA](#)


The County's public-education scoreboard: did a warning change a decision?

Beach checks by hour of day, island-wide

[SAMPLE DATA](#)


Decisions cluster **before 10am** — a hazard message sent at noon reaches an island that already chose.



THE DASHBOARD · CAPACITY AND DISPERSAL

Where the overflow actually goes.

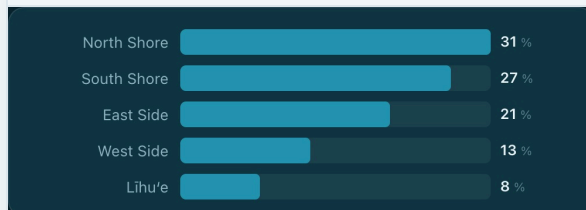
Hā'ena is capped and reserved, and the demand that misses a reservation does not evaporate — it lands on a north-shore lot with no gate and no count. Measured directly, that gap stops being an argument and becomes a number.

Kaua'i — visitor intent and lifeguard-tower coverage

[SAMPLE DATA](#)

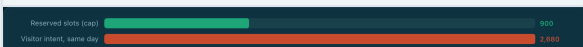

The whole island on one screen. **Circle size is visitor intent; red marks shoreline with no tower.** Coastline is the same outline the live app uses, so a pin sits where the beach actually is.

Visitor intent by district

[SAMPLE DATA](#)


Dispersal, measured — not surveyed after the fact.

North-shore overflow versus the cap

[SAMPLE DATA](#)


Roughly **three visitors of intent per reserved slot** on a peak sample day.



THE DASHBOARD · REEF PROTECTION

A reef signal, from a scanner already shipped.

Every beach page in the live app carries a sunscreen scanner. A visitor scans a barcode, the app reads the **active ingredients** rather than the marketing label, and returns Avoid, Caution, or Reef-safer against Hawai'i **Act 104** — which has banned the sale of oxybenzone and octinoxate sunscreen since January 2021. With visitor opt-in, those scans roll up into two County signals nothing else on the island produces.

Verdict mix across all scans

[SAMPLE DATA](#)


Roughly **a quarter of scanned bottles carry a filter Act 104 bans from sale** — the compliance picture the County has never had.

Signal 1 · Retail areas worth a spot-check

[SAMPLE DATA](#)


Share of Avoid-verdict scans **by area, never by store**. The suppressed row is the small-N guardrail working.

Signal 2 · Reef sampling priority

[SAMPLE DATA](#)


A **triage order for an ecological survey** when there is budget for three reefs, not thirty.



A signal to look, not evidence. Scanned is not sold — visitors arrive with mainland sunscreen. Scanned is not applied; a person can read Avoid and leave the bottle in the bag, which is the point of the feature. And density is not reef harm: warming seas are the dominant stressor, sunscreen filters an avoidable local one. **No retailer is ever named** — areas only, thresholded, aggregate.



THE DASHBOARD · ADVISORIES AND HONEST STATUS

Every advisory goes out unmeasured today.

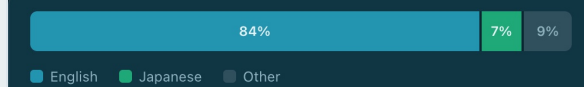
The County and DOH post brown-water advisories, high-surf warnings, and closures. Nobody currently measures how many visitors saw one before they got in the water. OceanSafe already carries these notices in-app, so reach and timing are countable per beach and per posting.

Advisory reach, by hours after posting

[SAMPLE DATA](#)


Reach is near-complete within six hours, then decays. **Posting time is the lever**, and it is measurable for the first time.

Session language, island-wide

[SAMPLE DATA](#)


Which languages the County can reach in an emergency without a translation vendor.

Counting today · In build · Needs County input

[SAMPLE DATA](#)

✓ Counting today

Beach checks by site and by hour

Guarded vs unguarded intent split

Advisory and hazard-verdict impressions

District-level distribution

Device type and session depth

The counting system is live and running today across 44 Kaua'i beaches — the County names which questions it wants answered; the system is already running.

Sample figures throughout. Real numbers scale with how many visitors reach the guide.

🕒 In build

Deflection pairs after a hazard verdict

Language and source-market mix

Lead time between check and arrival

Trail and hike intent

Overflow modeling against caps

🔗 Needs County input

Authoritative lifeguard tower roster

Incident records for validation

DOH advisory feed timing

Park capacity and reservation counts

Which questions matter most first

The status panel is reproduced verbatim from the dashboard — it names what is not finished as plainly as what is.



THE FULL CATALOGUE · 1 OF 2

Every dataset, ranked by what it is worth to Kaua'i.

Ordering is not alphabetical and not technical — it runs from the datasets that keep somebody alive down to the ones that make an annual report easier to write. The status on each row is honest.

Tier 1 · Life safety

OCEAN SAFETY · PARKS · PIO

1	Unguarded-beach intent index Which shoreline with no tower is drawing the most people this week?	COUNTING TODAY	4	Deflection rate Did the warning actually change the plan?	IN BUILD
2	Beach-level pre-arrival demand How many people are planning each beach, ranked, before they arrive?	COUNTING TODAY	5	Substitution pairs When a beach turns dangerous, which beach inherits the crowd?	IN BUILD
3	Hazard-verdict exposure How many people considered a beach while it was flagged dangerous?	COUNTING TODAY	6	Lead time to entry How many hours sit between the check and the swim — the warning window?	IN BUILD

Tier 2 · Crowd and capacity

OCEAN SAFETY · PARKS · BUDGET

7	Arrival clock by hour What hour does the island decide where to go?	COUNTING TODAY	11	Small-lot parking pressure Which beaches draw concentrated intent against tiny lots?	IN BUILD
8	Day-of-week and seasonal curve Which days and months carry the load, site by site?	COUNTING TODAY	12	Trail and hike intent Kalalau, Sleeping Giant, Waimea — a search-and-rescue pre-signal.	IN BUILD
9	District dispersal share Is visitor load actually spreading past Hanalei and Po'ipū?	COUNTING TODAY	13	Facility-gap ranking Which high-demand sites have no restroom, tower, or lot?	IN BUILD
10	Hā'ena overflow index Where does capped, unreserved north-shore demand land instead?	IN BUILD			

How to read the status column

COUNTING TODAY

Already collected by the running app. The County just has to ask for the report.

IN BUILD

The underlying events are captured; the analysis or reporting view is unfinished. Weeks, not quarters.

NEEDS COUNTY INPUT

Blocked on something only the County holds — a roster, an incident file, a capacity number, or a decision.



THE FULL CATALOGUE · 2 OF 2

Reef, movement, and the administrative layer.

Tier 3 · Reef, advisories, and environment

DOH LIAISON · PIO · DOH

14 Advisory reach per posting How many people actually saw the brown-water notice?	COUNTING TODAY	18 Act 104 verdict mix What share of scanned sunscreen carries a filter banned from sale in Hawai'i?	COUNTING TODAY
15 Advisory reach curve by hour How fast does a notice propagate — and when should it post?	COUNTING TODAY	19 Retail-area compliance triage Which retail AREAS are worth an Act 104 spot-check — never which store?	IN BUILD
16 Post-advisory behavior change Did stream-adjacent intent actually drop after the warning?	IN BUILD	20 Reef sampling priority Which reef should an ecological survey sample first, given a budget for three?	IN BUILD
17 Rain-keyed bacteria caution response Does the rain-triggered caution move anyone near canals and streams?	COUNTING TODAY	21 Chronic-hazard site list Which sites recur at the top of dangerous-conditions intent, year over year?	IN BUILD
		22 Shoreline-change exposure Which highest-demand beaches sit on eroding or sea-level-exposed shoreline?	NEEDS COUNTY INPUT

Tier 4 · Movement and economy

OED · PARKS · TRANSPORTATION

23 Weather-shift redistribution Where does the island go on a rainy day?	COUNTING TODAY	27 Resident versus visitor pattern How much of the load is resident, in aggregate proxy terms?	IN BUILD
24 Corridor directional pressure North versus south intent volume by hour along Kūhiō Highway.	IN BUILD	28 In-destination category intent Which town categories visitors reach for, by district.	COUNTING TODAY
25 Source-market mix Which markets are actually here — weekly, not quarterly.	IN BUILD	29 Trip-stage planning intensity Day one versus day five — when does the itinerary actually set?	IN BUILD
26 Language mix Which languages the County can reach directly, without a vendor.	IN BUILD		

Tier 5 · Operations, comms, and grants

EMERGENCY MGMT · PARKS · OCEAN SAFETY

30 Emergency-notice reach test How fast could a County message reach app users — measured, not assumed?	IN BUILD	32 Grant-evidence packet Before-and-after metrics formatted for federal and state applications.	IN BUILD
31 Physical sign and QR performance Which County signs and placards actually get scanned?	COUNTING TODAY	33 Public open-data dashboard A resident-facing page the County can publish as its own.	NEEDS COUNTY INPUT

What the County would owe in return

Three things, none of them money. The authoritative lifeguard tower roster, so the guarded-versus-unguarded split is correct rather than inferred. Historical incident locations, so the intent index can be validated against outcomes instead of asserted. And a named person who will say which of these questions matters first.



GOVERNANCE

The part that has to be right before anything else matters.

A visitor-data program attached to a county government fails in exactly one way, and it is worth naming plainly at the front rather than burying in a schedule. It fails when individual-level records exist somewhere and then move somewhere they should not. The design answer is not a policy promise. It is to never hold the records.

What OceanSafe holds

Counts and percentages, bucketed by beach, hour, district, and condition. A row in the system reads “Po‘ipū, 8am, 41 checks.” There is no name, no account, no device identifier, and no history that could be reassembled into a person’s day.

What that means practically

There is nothing individual to breach, subpoena, resell, or be accused of reselling. Small cells are suppressed rather than published. The County can state this to a reporter or a council member in one sentence and have it survive the follow-up questions.

Terms offered up front, before the County asks

- 1 Aggregate-only reporting.** No individual-level record is created, retained, or transferred, at any tier, for any party.
- 2 Perpetual free County access.** The County keeps free access to aggregates for its own jurisdiction, permanently, including if the commercial relationship ends.
- 3 Small-cell suppression.** Any bucket below a threshold is withheld rather than published, so no rare pattern can single anyone out.
- 4 Public methodology.** How a number is produced is documented and inspectable, so a figure in a council packet can be defended.
- 5 Safety content is never for sale.** The hazard verdict is not sponsorable, not placeable, and not influenced by any commercial arrangement, on any surface.
- 6 Exit without hostage.** The County can end the arrangement at any time and keep every aggregate it has already been given.



Aggregates are all that exist. OceanSafe does not store an individual person, device, or location history, so there is nothing individual to hand over, subpoena, breach, or sell. What the County receives is a count — the same shape as a traffic count or a park gate tally.

Offered as starting terms, not as a negotiating position. Counsel is welcome to tighten them.



WHERE THIS GOES

Safety first. Then the traffic goes to work for local business.

It is worth stating the plan plainly rather than letting the County find it out later. OceanSafe is free and will stay free, and the safety layer is the product on its own terms. But an app that a visitor opens before every beach day is also the most valuable thing a small Kaua'i business can be listed in — an audience that is already here, already deciding what to do next, and currently being routed to mainland booking sites and national chains. The second half of this plan is turning that attention toward locally owned businesses.

Phase one · now — visitor safety

A free conditions and hazard guide across 44 Kaua'i beaches, plus the reef scanner and the advisory layer. No charge to visitors, no charge to the County, and the full catalogue in this document handed over aggregate and free. This phase is already running.

Phase two · next — local commerce with built-in traffic

The same audience, in the app's Town tab: Kaua'i-owned shops, food, guides, and activities placed in front of visitors at the moment they are deciding — a one-van tour operator on the same shelf as a national brand, in the one place a visitor is already looking.

Why the County should want the second half, not just tolerate it

- 1 The money stays on the island.** Every booking that runs through a mainland platform exports value off Kaua'i. Routing the same visitor to a locally owned operator keeps it here — which is the entire remit of economic development.
- 2 Small operators get distribution they cannot otherwise buy.** A one-van tour company or a Hanapēpē shop cannot outbid a national brand for search placement. Placement inside an app the visitor already opened is a shelf they can actually reach.
- 3 Dispersal is already County policy.** Pulling visitors into town centers and the west side, on days the ocean is closed out, is the same goal the dispersal dataset measures — and commerce is the lever that actually moves people there.
- 4 It makes the safety layer permanent.** A free public-safety tool that depends on grants ends when the grant does. This one is built to sustain itself.



What does not change, in either phase. The safety verdict is never sponsorable, never placeable, and never influenced by who is paying — a business cannot buy its way onto a beach's conditions card, and no advertisement appears over a hazard warning. Commerce lives in a separate tab the visitor chooses to open. And the County's data stays aggregate, free, and permanent regardless of what the commercial side ever earns.



OceanSafe

WHAT WE ARE ASKING FOR

The data is the County's. We need help with the QR code.

Everything in this document — the full set of 33 datasets, aggregated, free, and permanent — is available to the County. There is no fee, no procurement, and nothing to negotiate. What OceanSafe needs in return is **distribution**. The app only produces a signal where visitors actually open it, and the County controls nearly every place a visitor stops on the way to the water. **The airport is the single highest-value placement** — it reaches a visitor before their first beach day, not after it — and we would take any surface the County or State is willing to allow.

At the shoreline

Ocean Safety towers, beach-park signage, restroom and pavilion boards — where people are already looking for the rules.

Before the beach

Līhu'e Airport arrivals, visitor centers, park kiosks, rental counters, and the County website — the hours when the decision is still open.

In County materials

Safety campaigns, advisory pages, and printed visitor guides — the same channels already carrying the message.



This is the code. Scan it.

It opens the free public guide — no account, no download, no advertisement over a safety verdict, ever. Placement can carry the County's own mark and wording. Three things would make the data sharper on day one: the authoritative lifeguard tower roster, historical incident locations for validation, and a named person who will say which of the 33 questions matters first.

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Tropical Agronomics dba OceanSafe · Kaua'i · The working dashboard in this document is available to view live on request.

Sample figures throughout. Real numbers begin as soon as the QR is in front of visitors and the County names the questions it wants counted.